# Broadband Data Collection

# Data Specifications for Mobile Speed Test Data

**March 31, 2025**

# Table of Contents

## Change Log

| Revision | Date | Comments |
|---|---|---|
| 1.0 | 2022-03-09 | Initial release. |
| 1.0.1 | 2022-03-17 | Added Change Log; updated spectrum_band field to record a string value instead of integer in Mobile Speed Test Data specification. |
| 1.1 | 2022-09-15 | Added Section 6 Entity Mobile Testing Methodology Information; updated text of Section 1 Overview. |
| 1.2 | 2022-11-08 | Fixed typos; updated data validation information throughout Section 5.1 to match BDC system validations. |
| 1.3 | 2022-11-18 | In Sections 2.1. and 2.1.2, clarified that, for bulk mobile crowdsource data, the listed Test Object attributes are optional if the test includes at least one download or upload speed measurement. |
| 2.0 | 2025-03-13 | Modified Section 5.1 to reflect newly-supported fields and updated examples; added Sections 5.1.8.1 and 5.1.8.2 to delineate two flavors of "cell" objects; added Section 5.2 to reflect newly-supported CSV data format. |
| 2.1 | 2025-03-31 | Modified arfcn field description in Section 5.1.8 to fix incorrect values. |

# 1 Overview

As part of the Federal Communication Commission's (FCC or the Commission) Broadband Data Collection (BDC), individuals and entities can take on-the-ground speed tests of mobile network coverage and submit the results of those tests to the FCC.

The speed test results for the BDC may be collected and submitted to the FCC via different mechanisms depending on who takes the speed test and the purpose of the test. Specifically, mobile speed test data fall into five categories:

1. Consumer Crowdsource: speed tests that are (a) conducted by members of the general public using the FCC Speed Test app or an approved third-party mobile app, and (b) are not intended to formally dispute a mobile service provider's claimed coverage. These test results are not considered valid challenge data and will be used by the Commission to inform its understanding of mobile coverage data.
2. Consumer Challenge: speed tests conducted by members of the general public using the FCC Speed Test app or an approved third-party app that are submitted to formally dispute a mobile service provider's claimed coverage.
3. Entity Crowdsource: speed tests that are (a) conducted by service providers, state, local, or Tribal governments, or other entities using their own hardware or software, and (b) are not submitted to formally dispute a mobile service provider's claimed coverage. These test results are not considered valid challenge data and will be used by the Commission to inform its understanding of mobile coverage data.
4. Entity Challenge: speed tests conducted by service providers, state, local, or Tribal governments, or other third-parties using their own hardware or software that are submitted to formally dispute a mobile service provider's claimed coverage.
5. Provider Response: speed tests that are conducted by service providers using their own hardware or software, and are submitted to provide on-the-ground measurements in response to a challenge or a Commission verification inquiry.

Consumers or members of the general public interested in submitting mobile speed tests as crowdsourced data or as part of the challenge process may use the FCC Speed Test app or other third-party apps approved by the Office of Engineering and Technology (OET) on iOS or Android devices to conduct and automatically submit data. All consumer speed test data must be transmitted to the FCC's BDC system by the third-party app developer from its servers via an Application Programming Interface (API), and such data must conform to the specifications detailed below.

Entities – including service providers, state, local, and Tribal governments, as well as other third-parties (e.g., consumer groups, non-governmental groups, etc.) – interested in submitting mobile speed tests as crowdsourced data or as part of the mobile challenge process may use their own hardware or software to conduct bulk collections of mobile speed tests. Such bulk speed test data must be submitted in the BDC system either via file upload or through use of the BDC filer API, and such data must conform to the specifications detailed below. An entity interested in submitting such data must first be authenticated in the BDC system.

Service providers that submit on-the-ground measurements in response to a challenge or a Commission verification inquiry may use their own hardware or software to conduct mobile speed tests. All provider response speed test data must be submitted in the BDC system either via file upload or through use of the BDC filer API, and such data must conform to the specifications detailed below. A service provider submitting such data must first be authenticated in the BDC system.

Section 5 of this document provides the data structure used for all mobile speed test data collected as part of the BDC. The data must be submitted in a JavaScript Object Notation (JSON) file format and match the specifications below. Sections 2 through 4 explain which attributes detailed in the common data structure are optional and which values may be null, depending on the type of entity submitting the data, the purpose of the data, the device operating system, and other factors. Section 6 describes the information that entities must provide when using their own hardware or software to collect on-the-ground speed test data.

Instructions for governmental entities, other entities, and mobile providers on how to use the API to submit data into the BDC system will be provided at a later date.

This document provides guidance on the requirements of the Commission's rules and explains how to make the required filings in the system. The rules governing the Broadband Data Collection (formerly known as the Digital Opportunity Data Collection) can be found in 47 CFR § 1.7004 *et seq.* Additional information on the rules can be found in the FCC's BDC orders, which are available at https://www.fcc.gov/BroadbandData/resources on the "Releases" tab.

The FCC may publish updates to this document. In addition, other materials related to the BDC generally, and mobile speed test data specifically, have been and will be made available on the FCC's Broadband Data Collection Resources page of the FCC's website at https://www.fcc.gov/BroadbandData/resources and at the online help center at http://www.fcc.gov/BroadbandData/help..

# 2 Crowdsourced Data

## 2.1 Mobile Crowdsourced Data

Mobile crowdsourced data is submitted by authenticated entities in the BDC system using either a file upload process (in the system's challenge process module) or a filer API. In addition, authorized mobile speed test app providers can submit consumer crowdsourced speed test results to the BDC system via an API. All mobile crowdsourced data must be submitted as tabular data file in a standardized format specified below.

Crowdsourced data may contain less information than required for mobile challenges, but otherwise will use the same data specifications and API specifications. Consumers and entities submitting crowdsourced data must provide identifying information and certify to the accuracy of the data.

| Data Name | Parties | Description / Notes |
|---|---|---|
| **Mobile Consumer Crowdsourced Data** | Consumers | Records of mobile speed tests or other mobile measurements (e.g., signal strength, etc.) in tabular format.<br><br>- *Note: providers will be notified when mobile consumer crowdsourced data have been submitted but such data do not, on their own, initiate the mobile challenge process or otherwise obligate a provider to respond.* |
| **Mobile Entity Crowdsourced Data** | Service Providers<br><br>Governmental Entities<br><br>Other Entities | Records of mobile speed tests or other mobile measurements (e.g., signal strength, etc.) in tabular format.<br><br>- *Note: providers will be notified when mobile entity crowdsourced data have been submitted but such data do not, on their own, initiate the mobile challenge process or otherwise obligate a provider to respond.* |

### 2.1.1 Mobile Consumer Crowdsourced Speed Test Data

This file must contain records of each mobile speed test. The file must be in JavaScript Object Notation (JSON) format and match the specifications provided in **Section 5.1**.

For crowdsourced data, the following **Test Object** attributes are optional so long as the test includes at least one download or upload speed measurement:

- download
- upload
- latency

To account for limitations on the iOS platform, the following attributes are optional or have values that may be null when submitting crowdsourced data for devices running iOS:

- **Submission Object**
  - device_tac
  - net_mobile_country_code
  - net_mobile_network_code
  - sim_mobile_country_code
  - sim_mobile_network_code
- **Download Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Upload Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Latency Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Cell Object**
  - physical_cell_id
  - cell_connection
  - signal_strength
  - rssi
  - rsrp
  - rsrq
  - sinr
  - csi_rsrp
  - csi_rsrq
  - csi_sinr
  - cqi
  - spectrum_band
  - spectrum_bandwidth
  - arfcn

### 2.1.2 Mobile Entity Crowdsourced Speed Test Data

This file contains records of each mobile speed test in JavaScript Object Notation (JSON) format matching the specifications provided in **Section 5.1**. All attributes and values are required (unless the attribute is nullable and the value is null).

For crowdsourced data, the following **Test Object** attributes are optional so long as the test includes at least one download or upload speed measurement:

- download
- upload
- latency

# 3 Challenge Process Data

## 3.1 Mobile Challenge Data

Speed test data for the BDC mobile challenge process can be submitted by authenticated entities in the BDC system using either a file upload process or a filer API. In addition, authorized mobile speed test app providers can submit mobile challenge speed test results from consumers into the BDC system via an API. All mobile crowdsourced data must be submitted as tabular data file in a standardized format specified below.

Challenge data submitted by consumers may be taken on iOS devices and therefore contain less information than is required for challenge data submitted by governmental or other entities, but in both cases challenge data must contain more information than required for crowdsourced data. Otherwise, the data will use the same data specifications and API specifications. Consumers submitting challenge data via mobile app must provide identifying information and certify to the accuracy of the data.

| Data Name | Parties | Description / Notes |
|---|---|---|
| **Mobile Consumer Challenge Speed Test Data** | Consumers | Records of mobile speed tests, including download, upload, and latency test metrics, challenging a provider's coverage data.<br><br>*- Note: certain technical measurements are unavailable on iOS devices and thus are optional for consumer challenge data.*<br><br>*- Note: providers will be notified when mobile challenge data have been submitted and a challenge created. Once an area is challenged, providers have 60 days to respond.* |
| **Mobile Entity Challenge Speed Test Data** | Service Providers<br><br>Governmental Entities<br><br>Other Entities | Records of mobile speed tests, including download, upload, and latency test metrics, challenging a provider's coverage data.<br><br>*- Note: all technical measurements are required for entity challenge data, and thus entities cannot use iOS devices to submit such data.*<br><br>*- Note: providers will be notified when mobile challenge data have been submitted and a challenge created. Once an area is challenged, providers have 60 days to respond.* |

### 3.1.1 Mobile Consumer Challenge Speed Test Data

This file must contain records of each mobile speed test. The file must be in JavaScript Object Notation (JSON) format and match the specifications provided in **Section 5.1**.

To account for limitations on the iOS platform, the following attributes are optional or have values that may be null when submitting challenge data for devices running iOS:

- **Submission Object**
  - device_tac
  - net_mobile_country_code
  - net_mobile_network_code
  - sim_mobile_country_code
  - sim_mobile_network_code
- **Download Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Upload Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Latency Test Object**
  - carrier_aggregation_flag
  - network_connected_flag
  - network_available_flag
  - network_roaming_flag
- **Cell Object**
  - physical_cell_id
  - cell_connection
  - signal_strength
  - rssi
  - rsrp
  - rsrq
  - sinr
  - csi_rsrp
  - csi_rsrq
  - csi_sinr
  - cqi
  - spectrum_band
  - spectrum_bandwidth
  - arfcn

### 3.1.2 Mobile Entity Challenge Speed Test Data

This file must contain records of each mobile speed test. The file must be in JavaScript Object Notation (JSON) format and match the specifications provided in **Section 5.1**. All attributes and values are required (unless the attribute is nullable and the value is null).

# 4 Provider Response Data

## 4.1 Mobile Provider Response Data

Mobile service providers whose coverage data have been challenged may optionally submit either the results of mobile speed tests or infrastructure information as response data, along with other data relevant to rebut the challenge, or else concede the challenge. Mobile service providers that are the subject of an inquiry as part of the Commission's verification process must submit either the results of mobile speed tests or infrastructure information. Regardless of the type of data submitted in response to a challenge or verification inquiry, the Commission may subsequently require the mobile service provider to submit additional information if needed to ensure an adequate review, including but not limited to either infrastructure or mobile speed test data (to the extent not the option initially chosen by the provider) or data collected from network transmitter monitoring systems or software (to the extent available in the provider's network).

Response speed test data submitted by mobile service providers must contain the same level of information required for challenge data submitted by governmental or other entities and must conform to the same data specifications and API specifications. The data must be submitted in the BDC system via file upload or an API. Mobile service providers submitting response data must provide certifications to the accuracy of the data.

| Data Name | Parties | Description / Notes |
|---|---|---|
| **Mobile Provider Response Test Data** | Service Providers | Records of mobile speed tests, including download, upload, and latency test metrics, measuring the provider's coverage for the area subject to challenge or verification inquiry in tabular format.<br>*- Note: providers have 60 days to submit response data.* |

### 4.1.1 Mobile Provider Response Speed Test Data

This file must contain records of each mobile speed test. The file must be in JavaScript Object Notation (JSON) format and match the specifications provided in **Section 5.1**. All attributes and values are required (unless the attribute is nullable and the value is null).

# 5 Common Data Formats

## 5.1 Hierarchical Mobile Speed Test Data (JSON)

This section details the data structure common for all mobile speed test data in the Broadband Data Collection. This file contains records of each mobile speed test in JavaScript Object Notation (JSON) format matching the specifications in the table and sections below:

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **submission_ category** | Enumerated | Consumer Challenge | Category of data submission.<br>*- Value must be one of the following:*<br>*{Consumer Crowdsourced\|Consumer Challenge\|Entity Crowdsourced\|Entity Challenge\|Provider Response\|Other}* |
| **contact** | Contact Object | | Contact information about the user.<br>*Note: the specifications for the Contact Object are described in **Section 5.1.1**.* |
| **submissions** | Array [Submission Object] | | List of crowdsourced data or challenge submissions.<br>*Note: the specifications for the Submission Object are described in **Section 5.1.2**. It includes the Test Object, which includes the Download Test Object, Upload Test Object, and Latency Test Object. Each of these contains at least two Location Objects and one or more Cell Objects.* |

### 5.1.1 Contact Object

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **name** | String | Jane Broadband | Full name of the user. |
| **email** | String | jane.broadband@fcc.gov | Email address of the user.<br>*- Value must match valid email address format, e.g.: atom@domain.tld.* |
| **phone** | String | 888-225-5322 | Phone number of the user.<br>*- Value must match valid US phone number format, if not null: 000-000-0000.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.* |

### 5.1.2 Submission Object

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **test_id** | String | 173422 | Unique identifier used by the app or entity to differentiate tests.<br>*- Value must be unique across all data submitted by a third-party speed test app vendor if the submission_category value is Consumer Crowdsourced or Consumer Challenge.*<br>*- Value must be unique across all data submitted by the same entity if the submission_category value is Entity Crowdsourced, Entity Challenge, Provider Response, or Other.* |
| **device_timestamp** | Datetime | 2024-09-06T17:33:38-04:00 | Timestamp of the time at which the test submission data were transmitted to the app's servers, measured by the device.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, e.g.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **server_timestamp** | Datetime | 2024-09-06T17:33:36-04:00 | Timestamp of the time at which the test submission data were transmitted to the app's servers, measured by the server.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset if not null, e.g.: YYYY-MM-DD[T]hh:mm:ss±hh:mm*<br>*- Value must correspond to transmission recorded in the server_source_ip_address and server_source_port values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **server_source_ ip_address** | String | 172.58.245.221 | Source IP address of the device submitting test submission data, measured by the server.<br>*- Value must be in valid IPv4 or IPv6 format if not null.*<br>*- Value must correspond to transmission recorded in the server_timestamp and server_source_port values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |
| **server_source_port** | Integer | 55758 | Source TCP port of the device submitting test submission data, measured by the server.<br>*- Value must correspond to transmission recorded in the server_timestamp and server_source_ip_address values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **device_imei** | String | | International Mobile Equipment Identity (IMEI) of the device.<br>*- Value must correspond to a valid 15-digit IMEI.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Consumer Challenger, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the server_timestamp, server_source_ip_address, and server_source_port values are not null.* |
| **device_type** | Enumerable | Android | Type of device.<br>*- Value must be one of the following:*<br>*{iOS\|Android\|Other}*<br>*- Value must be iOS or Android if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value must be Android or Other if the submission_category value is Entity Challenge or Provider Response.* |
| **manufacturer** | String | Google | Name of the device manufacturer. |
| **model** | String | Pixel 8 | Name of the device model |
| **operating_system** | String | 14 | Name and version of the device operating system. |
| **device_tac** | String | 35871196 | 8-digit Type Allocation Code of the device.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the app does not have requisite permissions, or the device does not return a valid value or else returns a value of unknown.* |
| **device_id** | String | 908cdafa-7fbe-4f9d-8ea4-03595ed6bba8 | Unique device or application installation identifier. |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **app_name** | String | FCC Speed Test App - Android | Name of the mobile speed test app.<br>*- Value is required and must match an approved App Name string for the third-party speed test app if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value may be null if the submission_category value is Entity Challenge, Entity Crowdsourced, Provider Response, or Other.* |
| **app_version** | String | 5.0.13 | Version of the mobile speed test app.<br>*- Value is required and must match an approved App Version string for the third-party speed test app if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value may be null if the submission_category value is Entity Challenge, Entity Crowdsourced, Provider Response, or Other.* |
| **provider_name** | String | Acme Wireless | Name of the mobile service provider. |
| **sim_mobile_ country_code** | String | 310 | Numeric string of the mobile service provider's mobile country code reported from the SIM card.<br>*- Value may be null when device is not connected to a network.* |
| **sim_mobile_ network_code** | String | 001 | Numeric string of the mobile service provider's mobile network code reported from the SIM card.<br>*- Value may be null when device is not connected to a network.* |
| **net_mobile_ country_code** | String | 310 | Numeric string of the mobile service provider's mobile country code reported from the connected network.<br>*- Value is not available on iOS and may be null for these device types.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **net_mobile_ network_code** | String | 001 | Numeric string of the mobile service provider's mobile country code reported from the connected network.<br><br>*- Value is not available on iOS and may be null for these device types.* |
| **in_vehicle_flag** | Boolean | false | Boolean flag indicating whether the test was conducted while in-vehicle or outdoors. |
| **environment_code** | Enumerated Integer | 0 | Code for the testing environment in which the test was conducted.<br><br>*- Value must be one of the following codes:*<br><br>*0 – Outdoor Stationary*<br>*1 – In-vehicle Mobile*<br>*2 – Indoors*<br><br>*- Value must be null if the in_vehicle_flag value is not null.* |
| **external_ antenna_flag** | Boolean | false | Boolean flag indicating whether an in-vehicle test was conducted using an antenna external to the vehicle.<br><br>*- Value must be false when in_vehicle_flag is false or environment_code is 0.*<br><br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Consumer Challenger, Provider Response, or Other.*<br><br>*- Value is required when the submission_category value is Entity Challenge or Entity Crowdsourced.* |
| **scheduled_test_flag** | Boolean | false | Boolean flag indicating whether the test was automated / scheduled or was user-initiated.<br><br>*- Value is required when the submission_category value is Consumer Crowdsourced or Entity Crowdsourced.* |
| **tests** | Test Object | | Information about the test metrics.<br><br>*- Note: the specifications for the Test Object are described in **Section 5.1.3**.* |

### 5.1.3 Test Object

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download** | Download Test Object | | Information about the download test metric.<br>*- Note: the specifications for the Download Test Object are described in **Section 5.1.4**.* |
| **upload** | Upload Test Object | | Information about the upload test metric.<br>*- Note: the specifications for the Download Test Object are described in **Section 5.1.5**.* |
| **latency** | Latency Test Object | | Information about the latency test metric.<br>*- Note: the specifications for the Download Test Object are described in **Section 5.1.6**.* |

### 5.1.4 Download Test Object

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **timestamp** | Datetime | 2024-09-06T17:33:20-04:00 | Timestamp of the time at which the connection for the test metric was initialized (i.e., prior to any warmup period during which the connection stabilized).<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **warmup_ duration** | Integer | 2997000 | Duration in microseconds that connection took to stabilize (e.g., TCP slow start) before the test metric commenced. |
| **warmup_bytes_ transferred** | Integer | 22970368 | Measured total amount of data in bytes that were transferred during the period the connection took to stabilize (e.g., TCP slow start) before the test metric commenced. |
| **duration** | Integer | 5005000 | Duration that the test metric took to complete in microseconds. |
| **bytes_transferred** | Integer | 107995136 | Measured total amount of data in bytes that the test metric transferred. |
| **bytes_sec** | Integer | 21573628 | Measure number of bytes per second that the test metric transferred. |
| **locations** | Array [Location Object] | | List of geographic coordinates of the locations measured during the speed test.<br>*- Note: the specifications for each Location Object element are described in **Section 5.1.7**.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **cells** | Array [Cell Object] | | List of cellular telephony information measured during the speed test.<br>*- Value may be null or an empty array if the test fails (i.e., success_flag and/or network_connected_flag is false).*<br>*- Note: the specifications for each Cell Object element are described in **Section 5.1.8**.* |
| **targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the test metric.<br>*- Value may be null or an empty array if the success_flag value for the Download Test object is false.*<br>*- Value is required when the success_flag and network_connected_flag values for the Download Test object is true.* |
| **connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>*- Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>*- Value may be null and will be treated as "cell".* |
| **success_flag** | Boolean | true | Boolean flag indicating whether the test completed successfully and without a change in state or connectivity. |
| **carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected.<br>*- Value is not available on iOS and may be null for these device types.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available.<br>*- Value is not available on iOS and may be null for these device types.* |
| **network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming.<br>*- Value is not available on iOS and may be null for these device types.* |

### 5.1.5 Upload Test Object

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **timestamp** | Datetime | 2024-09-06T17:33:28-04:00 | Timestamp of the time at which the connection for the test metric was initialized (i.e., prior to any warmup period during which the connection stabilized).<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **warmup_ duration** | Integer | 3688000 | Duration in microseconds that connection took to stabilize (e.g., TCP slow start) before the test metric commenced. |
| **warmup_bytes_ transferred** | Integer | 7864320 | Measured total amount of data in bytes that were transferred during the period the connection took to stabilize (e.g., TCP slow start) before the test metric commenced. |
| **duration** | Integer | 5343000 | Duration that the test metric took to complete in microseconds. |
| **bytes_transferred** | Integer | 14157824 | Measured total amount of data in bytes that the test metric transferred. |
| **bytes_sec** | Integer | 2649672 | Measure number of bytes per second that the test metric transferred. |
| **locations** | Array [Location Object] | | List of geographic coordinates of the locations measured during the speed test.<br>*- Note: the specifications for each Location Object element are described in **Section 5.1.7**.* |
| **cells** | Array [Cell Object] | | List of cellular telephony information measured during the speed test.<br>*- Value may be null or an empty array if the test fails (i.e., success_flag is false).*<br>*- Note: the specifications for each Cell Object element are described in **Section 5.1.8**.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the test metric.<br>*- Value may be null or an empty array if the success_flag value for the Upload Test object is false.*<br>*- Value is required when the success_flag and network_connected_flag values for the Upload Test object is true.* |
| **connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>*- Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>*- Value may be null and will be treated as "cell".* |
| **success_flag** | Boolean | true | Boolean flag indicating whether the test completed successfully and without a change in state or connectivity. |
| **carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected.<br>*- Value is not available on iOS and may be null for these device types.* |
| **network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available.<br>*- Value is not available on iOS and may be null for these device types.* |
| **network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming.<br>*- Value is not available on iOS and may be null for these device types.* |

### 5.1.6 Latency Test Object

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the test metric commenced.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **duration** | Integer | 2490000 | Duration that the test metric took to complete in microseconds. |
| **round_trip_time** | Integer | 30000 | Round-trip latency in microseconds. |
| **jitter** | Integer | 8000 | Round-trip jitter in microseconds. |
| **packets_sent** | Integer | 200 | Number of packets sent during the test. |
| **packets_received** | Integer | 200 | Number of packets received during the test. |
| **locations** | Array [Location Object] | | List of geographic coordinates of the locations measured during the speed test.<br>*- Note: the specifications for each Location Object element are described in **Section 5.1.7**.* |
| **cells** | Array [Cell Object] | | List of cellular telephony information measured during the speed test.<br>*- Value may be null or an empty array if the test fails (i.e., success_flag is false).*<br>*- Note: the specifications for each Cell Object element are described in **Section 5.1.8**.* |
| **targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the test metric.<br>*- Value may be null or an empty array if the success_flag value for the Latency Test object is false.*<br>*- Value is required when the success_flag and network_connected_flag values for the Latency Test object is true.* |
| **connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>*- Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>*- Value may be null and will be treated as "cell".* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **success_flag** | Boolean | true | Boolean flag indicating whether the test completed successfully and without a change in state or connectivity. |
| **carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected.<br>*- Value is not available on iOS and may be null for these device types.* |
| **network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available.<br>*- Value is not available on iOS and may be null for these device types.* |
| **network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming.<br>*- Value is not available on iOS and may be null for these device types.* |

### 5.1.7 Location Objects

Each element of the "locations" array contains the geographic coordinates of the locations measured at the start and end of the speed test, as well as during the test (if measured).

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix).<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted.<br>*- Value must have minimum precision of 6 decimal digits.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted.<br>*- Value must have minimum precision of 6 decimal digits.* |
| **horizontal_ accuracy** | Numeric | 12.34840 | Horizontal accuracy of the location, radial, in meters measured from the device.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **speed** | Numeric | 0 | Speed in meters per second measured from the device.<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **speed_accuracy** | Numeric | 20 | Speed accuracy in meters per second measured from the device.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

### 5.1.8 Cell Objects

The format of the objects in the "cells" array differs slightly between standard BDC-formatted mobile speed test data and USF-formatted mobile speed test data (e.g., submitted in response to a USF verification request).

#### 5.1.8.1 Cell Objects (Standard)

Each element of the "cells" array contains telephony information about the cell / carrier for standard BDC-formatted data.

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the cell information was measured.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **cell_id** | Numeric | 10283271 | Measured mobile broadcast cell identifier.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **physical_cell_id** | Integer | 90 | Measured Physical Cell Identity (PCI) of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **cell_connection** | Enumerated | 2 | Connection status of the cell.<br>*- Value must be one of the following codes:*<br>*0 – Not Serving*<br>*1 – Primary Serving*<br>*2 – Secondary Serving*<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **network_ generation** | Enumerated | 5G | String representing the network generation of the cell.<br>*- Value must be one of the following:*<br>*{2G\|3G\|4G\|5G\|Other}* |
| **network_subtype** | Enumerated | NRNSA | String representing the network subtype of the cell.<br>*- Value must be one of the following, if not null:*<br>*{1X\|EVDO\|WCDMA\|GSM\|HSPA\|HSPA+\| LTE\|NRSA\|NRNSA}*<br>*- Note: this value must be NRSA when the cell is in standalone mode and NRNSA when the test is in non-standalone mode for 5G-NR tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **signal_strength** | Integer | -96 | Measured signal strength in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Received Signal Strength Indication (RSSI) for 3G tests or the Reference Signal Received Power (RSRP) for 4G LTE or 5G-NR tests.* |
| **rssi** | Integer | | Measured Received Signal Strength Indication (RSSI) in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 5G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **rsrp** | Integer | -96 | Measured Reference Signal Received Power (RSRP) in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **rsrq** | Integer | -11 | Measured Reference Signal Received Quality (RSRQ) in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **sinr** | Integer | 1 | Measured Signal to Interference and Noise Ratio (SINR) in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **csi_rsrp** | Integer | | Measured 5G Channel State Information (CSI) RSRP in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **csi_rsrq** | Integer | | Measured Channel State Information (CSI) RSRQ in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **csi_sinr** | Integer | | Measured Channel State Information (CSI) SINR in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **cqi** | Integer | 9 | Measured Channel Quality Indicator (CQI) of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **spectrum_band** | String | n2 | Spectrum band used by the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: the reported band value corresponds to the Operating Bands tables as follows:*<br>*- 4G LTE: 3GPP TS 36.101 section 5.5*<br>*- 5G-NR: 3GPP TS 38.101 table 5.2-1* |
| **spectrum_ bandwidth** | Numeric | 15 | Total amount of spectral bandwidth used by the cell in MHz.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **arfcn** | Integer | 675 | Measured physical RF channel number of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

#### 5.1.8.2 Cell Objects (USF)

Each element of the "cells" array contains telephony information about the cell / carrier for USF-formatted data.

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the cell information was measured.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **cell_id** | Numeric | 10283271 | Measured mobile broadcast cell identifier.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **physical_cell_id** | Integer | 90 | Measured Physical Cell Identity (PCI) of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **cell_connection** | Enumerated | 2 | Connection status of the cell.<br>*- Value must be one of the following codes:*<br>*0 – Not Serving*<br>*1 – Primary Serving*<br>*2 – Secondary Serving*<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **network_ generation** | Enumerated | 5G | String representing the network generation of the cell.<br>*- Value must be one of the following:*<br>*{2G\|3G\|4G\|5G\|Other}* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **network_subtype** | Enumerated | NRNSA | String representing the network subtype of the cell.<br>*- Value must be one of the following, if not null:*<br>*{1X\|EVDO\|WCDMA\|GSM\|HSPA\|HSPA+\|LTE\|NRSA\|NRNSA}*<br>*- Note: this value must be NRSA when the cell is in standalone mode and NRNSA when the test is in non-standalone mode for 5G-NR tests.* |
| **signal_strength** | Integer | -96 | Measured signal strength in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Received Signal Strength Indication (RSSI) for 3G tests or the Reference Signal Received Power (RSRP) for 4G LTE or 5G-NR tests.* |
| **rssi** | Integer | | Measured Received Signal Strength Indication (RSSI) in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 5G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **rxlev** | Decimal | | Measured Received Signal Level in dBm of the cell.<br>*- Value is only required for tests with a network generation of 2G and network subtype of GSM, and must be null for all other network generations or network subtypes.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **rsrp** | Integer | -96 | Measured Reference Signal Received Power (RSRP) in dBm of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **rsrq** | Integer | -11 | Measured Reference Signal Received Quality (RSRQ) in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **sinr** | Integer | 1 | Measured Signal to Interference and Noise Ratio (SINR) in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **rxqual** | Integer | | Measured Received Signal Quality of the cell<br><br>*- Value must be between* 0 and 7.<br><br>*- Value is only required for tests with a network generation of 2G and network subtype of GSM, and must be null for all other network generations or network subtypes.* |
| **ec_io** | Decimal | | Measured Energy per Chip to Interference Power Ratio in dB of the cell.<br><br>*- Value is only required for CDMA 1X, EVDO, WCDMA, HSPA, and HSPA+ network subtypes, and must be null for all other network subtypes.* |
| **rscp** | Decimal | | Measured Received Signal Code Power in dBm of the cell.<br><br>*- Value is only required for WCDMA, HSPA, and HSPA+ network subtypes, and may be null for all other network subtypes.* |
| **csi_rsrp** | Integer | | Measured 5G Channel State Information (CSI) RSRP in dBm of the cell.<br><br>*- Value is not available on iOS and may be null for these device types.*<br><br>*- Value must be null for 3G or 4G LTE tests.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **csi_rsrq** | Integer | | Measured Channel State Information (CSI) RSRQ in dB of the cell.<br><br>*- Value is not available on iOS and may be null for these device types.*<br><br>*- Value must be null for 3G or 4G LTE tests.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **csi_sinr** | Integer | | Measured Channel State Information (CSI) SINR in dB of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **cqi** | Integer | 9 | Measured Channel Quality Indicator (CQI) of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **spectrum_band** | String | n2 | Spectrum band used by the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: the reported band value corresponds to the Operating Bands tables as follows:*<br>*- 4G LTE: 3GPP TS 36.101 section 5.5*<br>*- 5G-NR: 3GPP TS 38.101 table 5.2-1* |
| **spectrum_ bandwidth** | Numeric | 15 | Total amount of spectral bandwidth used by the cell in MHz.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **arfcn** | Integer | 675 | Measured physical RF channel number of the cell.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

## 5.2 Tabular Mobile Speed Test Data (CSV)

This section details the data structure common for all mobile speed test data in the Broadband Data Collection when submitted in flattened / tabular format. This file contains records of each mobile speed test in Comma Separated Values (CSV) format matching the specification in the table below:

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **submission_ category** | Enumerated | Consumer Challenge | Category of data submission.<br>*- Value must be one of the following:*<br>*{Consumer Crowdsourced\|Consumer Challenge\|Entity Crowdsourced\|Entity Challenge\|Provider Response\|Other}* |
| **contact_name** | String | Jane Broadband | Full name of the user. |
| **contact_email** | String | jane.broadband@fcc.gov | Email address of the user.<br>*- Value must match valid email address format, e.g.: atom@domain.tld.* |
| **contact_phone** | String | 888-225-5322 | Phone number of the user.<br>*- Value must match valid US phone number format: 000-000-0000.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **test_id** | String | 173422 | Unique identifier used by the app or entity to differentiate tests.<br>*- Value must be unique across all data submitted by a third-party speed test app vendor if the submission_category value is Consumer Crowdsourced or Consumer Challenge.*<br>*- Value must be unique across all data submitted by the same entity if the submission_category value is Entity Crowdsourced, Entity Challenge, Provider Response, or Other.* |
| **device_timestamp** | Datetime | 2024-09-06T17:33:38-04:00 | Timestamp of the time at which the test submission data were transmitted to the app's servers, measured by the device.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, e.g.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **server_timestamp** | Datetime | 2024-09-06T17:33:36-04:00 | Timestamp of the time at which the test submission data were transmitted to the app's servers, measured by the server.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset if not null, e.g.: YYYY-MM-DD[T]hh:mm:ss±hh:mm*<br>*- Value must correspond to transmission recorded in the server_source_ip_address and server_source_port values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **server_source_ip_address** | String | 172.58.245.221 | Source IP address of the device submitting test submission data, measured by the server.<br>*- Value must be in valid IPv4 or IPv6 format if not null.*<br>*- Value must correspond to transmission recorded in the server_timestamp and server_source_port values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |
| **server_source_port** | Integer | 55758 | Source TCP port of the device submitting test submission data, measured by the server.<br>*- Value must correspond to transmission recorded in the server_timestamp and server_source_ip_address values as measured by the server, if not null.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the device_imei value is not null.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **device_imei** | String | | International Mobile Equipment Identity (IMEI) of the device.<br>*- Value must correspond to a valid 15-digit IMEI.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, Consumer Challenge, Provider Response, or Other.*<br>*- Value may be null if the submission_category value is Entity Challenge and the server_timestamp, server_source_ip_address, and server_source_port values are not null.* |
| **device_type** | Enumerable | Android | Type of device.<br>*- Value must be one of the following:*<br>*{iOS\|Android\|Other}*<br>*- Value must be iOS or Android if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value must be Android or Other if the submission_category value is Entity Challenge or Provider Response.* |
| **manufacturer** | String | Google | Name of the device manufacturer. |
| **model** | String | Pixel 8 | Name of the device model |
| **operating_system** | String | 14 | Name and version of the device operating system. |
| **device_tac** | String | 35871196 | 8-digit Type Allocation Code of the device.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the app does not have requisite permissions, or the device does not return a valid value or else returns a value of unknown.* |
| **device_id** | String | 908cdafa-7fbe-4f9d-8ea4-03595ed6bba8 | Unique device or application installation identifier. |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **app_name** | String | FCC Speed Test App - Android | Name of the mobile speed test app.<br>*- Value is required and must match an approved App Name string for the third-party speed test app if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value may be null if the submission_category value is Entity Challenge, Entity Crowdsourced, Provider Response, or Other.* |
| **app_version** | String | 5.0.13 | Version of the mobile speed test app.<br>*- Value is required and must match an approved App Version string for the third-party speed test app if the submission_category value is Consumer Challenge or Consumer Crowdsourced.*<br>*- Value may be null if the submission_category value is Entity Challenge, Entity Crowdsourced, Provider Response, or Other.* |
| **provider_name** | String | T-Mobile | Name of the mobile service provider. |
| **sim_mobile_ country_code** | String | 310 | Numeric string of the mobile service provider's mobile country code reported from the SIM card.<br>*- Value may be null when device is not connected to a network.* |
| **sim_mobile_ network_code** | String | 260 | Numeric string of the mobile service provider's mobile network code reported from the SIM card.<br>*- Value may be null when device is not connected to a network.* |
| **net_mobile_ country_code** | String | 310 | Numeric string of the mobile service provider's mobile country code reported from the connected network.<br>*- Value is not available on iOS and may be null for these device types.* |
| **net_mobile_ network_code** | String | 260 | Numeric string of the mobile service provider's mobile country code reported from the connected network.<br>*- Value is not available on iOS and may be null for these device types.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **in_vehicle_flag** | Boolean | false | Boolean flag indicating whether the test was conducted while in-vehicle or outdoors. |
| **environment_code** | Enumerable | 0 | |
| **external_ antenna_flag** | Boolean | false | Boolean flag indicating whether an in-vehicle test was conducted using an antenna external to the vehicle.<br>*- Value must be false when in_vehicle_flag is false.*<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Consumer Challenger, Provider Response, or Other.*<br>*- Value is required when the submission_category value is Entity Challenge or Entity Crowdsourced.* |
| **scheduled_test_flag** | Boolean | false | Boolean flag indicating whether the test was automated / scheduled or was user-initiated.<br>*- Value is required when the submission_category value is Consumer Crowdsourced or Entity Crowdsourced.* |
| **download_ timestamp** | Datetime | 2024-09-06T17:33:28-04:00 | Timestamp of the time at which the connection for the download test metric was initialized (i.e., prior to any warmup period during which the connection stabilized).<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **download_ warmup_ duration** | Integer | 3688000 | Duration in microseconds that connection took to stabilize (e.g., TCP slow start) before the download test metric commenced. |
| **download_ warmup_bytes_ transferred** | Integer | 7864320 | Measured total amount of data in bytes that were transferred during the period the connection took to stabilize (e.g., TCP slow start) before the download test metric commenced. |
| **download_ duration** | Integer | 5343000 | Duration in microseconds that the download test metric took to complete. |
| **download_ bytes_transferred** | Integer | 14157824 | Measured total amount of data in bytes that the download test metric transferred. |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **download_ bytes_sec** | Integer | 2649672 | Measured number of bytes per second that the download test metric transferred. |
| **download_ targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the download test metric.<br>*- Value may be null or an empty array if the success_flag value for the Download Test object is false.*<br>*- Value is required when the success_flag value for the Download Test object is true.* |
| **download_ connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>*- Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>*- Value may be null and will be treated as "cell".* |
| **download_ success_flag** | Boolean | true | Boolean flag indicating whether the download test completed successfully and without a change in state or connectivity. |
| **download_ carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.* |
| **download_ network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download_ network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming for the download test metric.<br><br>*- Value is not available on iOS and may be null for these device types.* |
| **download_ location_start_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the start of the download test metric.<br><br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **download_ location_start_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the start of the download test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **download_ location_start_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the start of the download test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **download_ location_start_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the start of the download test metric.<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ location_start_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the start of the download test metric.<br><br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download_ location_start_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the start of the download test metric.<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ location_end_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the end of the download test metric.<br><br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **download_ location_end_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the end of the download test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **download_ location_end_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the end of the download test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **download_ location_end_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the end of the download test metric.<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ location_end_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the end of the download test metric.<br><br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **download_ location_end_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the end of the download test metric.<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the cell information was measured for the primary cell for the download test metric.<br><br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **download_ cell_primary_ cell_id** | Numeric | 10283271 | Measured mobile broadcast cell identifier for the primary cell for the download test metric.<br><br>*- Value is not available on iOS and may be null for these device types.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ physical_cell_id** | Integer | 90 | Measured Physical Cell Identity (PCI) of the primary cell for the download test metric.<br><br>*- Value is not available on iOS and may be null for these device types.*<br><br>*- Value must be null for 3G tests.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ cell_connection** | Enumerated | 1 | Connection status of the primary cell for the download test metric.<br><br>*- Value must be one of the following codes:*<br><br>*0 – Not Serving*<br>*1 – Primary Serving*<br>*2 – Secondary Serving*<br><br>*- Value is not available on iOS and may be null for these device types.*<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download_ cell_primary_ network_ generation** | Enumerated | 5G | String representing the network generation of the primary cell for the download test metric.<br>*- Value must be one of the following:*<br>*{2G\|3G\|4G\|5G\|Other}* |
| **download_ cell_primary_ network_subtype** | Enumerated | NRNSA | String representing the network subtype of the primary cell for the download test metric.<br>*- Value must be one of the following:*<br>*{1X\|EVDO\|WCDMA\|GSM\|HSPA\|HSPA+\| LTE\|NRSA\|NRNSA}*<br>*- Note: this value must be NRSA when the cell is in standalone mode and NRNSA when the test is in non-standalone mode for 5G-NR tests.* |
| **download_ cell_primary_ signal_strength** | Integer | -96 | Measured signal strength in dBm of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Received Signal Strength Indication (RSSI) for 3G tests or the Reference Signal Received Power (RSRP) for 4G LTE or 5G-NR tests.* |
| **download_ cell_primary_ rssi** | Integer | | Measured Received Signal Strength Indication (RSSI) in dBm of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 5G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **download_ cell_primary_ rsrp** | Integer | -96 | Measured Reference Signal Received Power (RSRP) in dBm of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **download_ cell_primary_ rsrq** | Integer | -11 | Measured Reference Signal Received Quality (RSRQ) in dB of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **download_ cell_primary_ sinr** | Integer | 1 | Measured Signal to Interference and Noise Ratio (SINR) in dB of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download_ cell_primary_ csi_rsrp** | Integer | | Measured 5G Channel State Information (CSI) RSRP in dBm of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ csi_rsrq** | Integer | | Measured Channel State Information (CSI) RSRQ in dB of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ csi_sinr** | Integer | | Measured Channel State Information (CSI) SINR in dB of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ cqi** | Integer | 9 | Measured Channel Quality Indicator (CQI) of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **download_ cell_primary_ spectrum_band** | String | n2 | Spectrum band used by the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: the reported band value corresponds to the Operating Bands tables as follows:*<br>*- 4G LTE: 3GPP TS 36.101 section 5.5*<br>*- 5G-NR: 3GPP TS 38.101 table 5.2-1* |
| **download_ cell_primary_ spectrum_ bandwidth** | Numeric | 15 | Total amount of spectral bandwidth used by the primary cell for the download test metric in MHz.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **download_ cell_primary_ arfcn** | Integer | 675 | Measured physical RF channel number of the primary cell for the download test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ timestamp** | Datetime | 2024-09-06T17:33:28-04:00 | Timestamp of the time at which the connection for the upload test metric was initialized (i.e., prior to any warmup period during which the connection stabilized).<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **upload_ warmup_ duration** | Integer | 3688000 | Duration in microseconds that connection took to stabilize (e.g., TCP slow start) before the upload test metric commenced. |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ warmup_bytes_ transferred** | Integer | 7864320 | Measured total amount of data in bytes that were transferred during the period the connection took to stabilize (e.g., TCP slow start) before the upload test metric commenced. |
| **upload_ duration** | Integer | 5343000 | Duration in microseconds that the upload test metric took to complete. |
| **upload_ bytes_transferred** | Integer | 14157824 | Measured total amount of data in bytes that the upload test metric transferred. |
| **upload_ bytes_sec** | Integer | 2649672 | Measured number of bytes per second that the upload test metric transferred. |
| **upload_ targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the upload test metric.<br>*- Value may be null or an empty array if the success_flag value for the Upload Test object is false.*<br>*- Value is required when the success_flag value for the Upload Test object is true.* |
| **upload_ connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>*- Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>*- Value may be null and will be treated as "cell".* |
| **upload_ success_flag** | Boolean | true | Boolean flag indicating whether the upload test completed successfully and without a change in state or connectivity. |
| **upload_ carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected for the upload test metric.<br><br>*- Value is not available on iOS and may be null for these device types.* |
| **upload_ network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available for the upload test metric.<br><br>*- Value is not available on iOS and may be null for these device types.* |
| **upload_ network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming for the upload test metric.<br><br>*- Value is not available on iOS and may be null for these device types.* |
| **upload_ location_start_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the start of the upload test metric.<br><br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **upload_ location_start_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the start of the upload test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **upload_ location_start_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the start of the upload test metric.<br><br>*- Value must have minimum precision of 6 decimal digits.* |
| **upload_ location_start_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the start of the upload test metric.<br><br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ location_start_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the start of the upload test metric.<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ location_start_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the start of the upload test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ location_end_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the end of the upload test metric.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **upload_ location_end_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the end of the upload test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |
| **upload_ location_end_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the end of the upload test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |
| **upload_ location_end_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the end of the upload test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ location_end_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the end of the upload test metric.<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ location_end_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the end of the upload test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ timestamp** | Datetime | 2022-01-28T10:43:57-05:00 | Timestamp of the time at which the cell information was measured for the primary cell for the upload test metric.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **upload_ cell_primary_ cell_id** | Numeric | 33275650 | Measured mobile broadcast cell identifier for the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ physical_cell_id** | Integer | 391 | Measured Physical Cell Identity (PCI) of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **upload_ cell_primary_ cell_connection** | Enumerated | 2 | Connection status of the primary cell for the upload test metric.<br>*- Value must be one of the following codes:*<br>*0 – Not Serving*<br>*1 – Primary Serving*<br>*2 – Secondary Serving*<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ network_ generation** | Enumerated | 5G | String representing the network generation of the primary cell for the upload test metric.<br>*- Value must be one of the following:*<br>*{2G\|3G\|4G\|5G\|Other}* |
| **upload_ cell_primary_ network_subtype** | Enumerated | NRNSA | String representing the network subtype of the primary cell for the upload test metric.<br>*- Value must be one of the following:*<br>*{1X\|EVDO\|WCDMA\|GSM\|HSPA\|HSPA+\|LTE\|NRSA\|NRNSA}*<br>*- Note: this value must be NRSA when the cell is in standalone mode and NRNSA when the test is in non-standalone mode for 5G-NR tests.* |
| **upload_ cell_primary_ signal_strength** | Integer | -103 | Measured signal strength in dBm of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Received Signal Strength Indication (RSSI) for 3G tests or the Reference Signal Received Power (RSRP) for 4G LTE or 5G-NR tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ cell_primary_ rssi** | Integer | | Measured Received Signal Strength Indication (RSSI) in dBm of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 5G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ rsrp** | Integer | -103 | Measured Reference Signal Received Power (RSRP) in dBm of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **upload_ cell_primary_ rsrq** | Integer | -12 | Measured Reference Signal Received Quality (RSRQ) in dB of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ cell_primary_ sinr** | Integer | 2 | Measured Signal to Interference and Noise Ratio (SINR) in dB of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **upload_ cell_primary_ csi_rsrp** | Integer | -103 | Measured 5G Channel State Information (CSI) RSRP in dBm of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ csi_rsrq** | Integer | -12 | Measured Channel State Information (CSI) RSRQ in dB of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ csi_sinr** | Integer | 2 | Measured Channel State Information (CSI) SINR in dB of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **upload_ cell_primary_ cqi** | Integer | 4 | Measured Channel Quality Indicator (CQI) of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ spectrum_band** | String | n41 | Spectrum band used by the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: the reported band value corresponds to the Operating Bands tables as follows:*<br>*- 4G LTE: 3GPP TS 36.101 section 5.5*<br>*- 5G-NR: 3GPP TS 38.101 table 5.2-1* |
| **upload_ cell_primary_ spectrum_ bandwidth** | Numeric | 100.0 | Total amount of spectral bandwidth used by the primary cell for the upload test metric in MHz.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **upload_ cell_primary_ arfcn** | Integer | 528000 | Measured physical RF channel number of the primary cell for the upload test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ timestamp** | Datetime | 2024-09-06T17:33:14-04:00 | Timestamp of the time at which the latency test metric commenced.<br>- *Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **latency_ duration** | Integer | 2490000 | Duration that the latency test metric took to complete in microseconds. |
| **latency_ round_trip_time** | Integer | 30000 | Round-trip latency in microseconds. |
| **latency_ jitter** | Integer | 8000 | Round-trip jitter in microseconds. |
| **latency_ packets_sent** | Integer | 200 | Number of packets sent during the latency test. |
| **latency_ packets_received** | Integer | 200 | Number of packets received during the latency test. |
| **latency_ targets** | Array [String] | [ "s1-fcc-north-carolina.mozark.ai" ] | List of hostname or IP address of target server(s) used for the latency test metric.<br>- *Value may be null or an empty array if the success_flag value for the Latency Test object is false.*<br>- *Value is required when the success_flag value for the Latency Test object is true.* |
| **latency_ connection_type** | Enumerated String | cell | Identifier for the connection type (i.e., mobile cellular, WiFi, or other) under which the test was conducted.<br>- *Value must be one of the following codes, if not null:*<br>*{cell\|wifi\|other}*<br>- *Value may be null and will be treated as "cell".* |
| **latency_ success_flag** | Boolean | true | Boolean flag indicating whether the latency test completed successfully and without a change in state or connectivity. |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ carrier_ aggregation_flag** | Boolean | true | Boolean flag indicating whether the network used carrier aggregation during the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ network_ connected_flag** | Boolean | true | Boolean flag indicating whether the network is connected for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.* |
| **latency_ network_ available_flag** | Boolean | true | Boolean flag indicating whether the network is available for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.* |
| **latency_ network_ roaming_flag** | Boolean | false | Boolean flag indicating whether the network is roaming for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.* |
| **latency_ location_start_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the start of the latency test metric.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **latency_ location_start_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the start of the latency test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ location_start_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the start of the latency test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |
| **latency_ location_start_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the start of the latency test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ location_start_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the start of the latency test metric.<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ location_start_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the start of the latency test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ location_end_ timestamp** | Datetime | 2022-01-28T10:44:07-05:00 | Timestamp of the time at which the location was determined (i.e., time of this location fix) at the end of the latency test metric.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **latency_ location_end_ latitude** | Decimal | 38.8860976 | Unprojected (WGS-84) geographic coordinate latitude in decimal degrees of the reported location where the test was conducted at the end of the latency test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ location_end_ longitude** | Decimal | -76.9962846 | Unprojected (WGS-84) geographic coordinate longitude in decimal degrees of the reported location where the test was conducted at the end of the latency test metric.<br>*- Value must have minimum precision of 6 decimal digits.* |
| **latency_ location_end_ horizontal_ accuracy** | Numeric | 2.0 | Horizontal accuracy of the location, radial, in meters measured from the device at the end of the latency test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ location_end_ speed** | Numeric | 0 | Speed in meters per second measured from the device at the end of the latency test metric.<br>*- Value may be null if the submission_category value is Consumer Crowdsourced, Entity Crowdsourced, or Other.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ location_end_ speed_accuracy** | Numeric | 1.0 | Speed accuracy in meters per second measured from the device at the end of the latency test metric.<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ timestamp** | Datetime | 2022-01-28T10:43:57-05:00 | Timestamp of the time at which the cell information was measured for the primary cell for the latency test metric.<br>*- Value must match valid ISO-8601 format including seconds and timezone offset, i.e.: YYYY-MM-DD[T]hh:mm:ss±hh:mm* |
| **latency_ cell_primary_ cell_id** | Numeric | 33275650 | Measured mobile broadcast cell identifier for the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ cell_primary_ physical_cell_id** | Integer | 391 | Measured Physical Cell Identity (PCI) of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ cell_connection** | Enumerated | 2 | Connection status of the primary cell for the latency test metric.<br>*- Value must be one of the following codes:*<br>*0 – Not Serving*<br>*1 – Primary Serving*<br>*2 – Secondary Serving*<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ network_ generation** | Enumerated | 5G | String representing the network generation of the primary cell for the latency test metric.<br>*- Value must be one of the following:*<br>*{2G\|3G\|4G\|5G\|Other}* |
| **latency_ cell_primary_ network_subtype** | Enumerated | NRNSA | String representing the network subtype of the primary cell for the latency test metric.<br>*- Value must be one of the following:*<br>*{1X\|EVDO\|WCDMA\|GSM\|HSPA\|HSPA+\| LTE\|NRSA\|NRNSA}*<br>*- Note: this value must be NRSA when the cell is in standalone mode and NRNSA when the test is in non-standalone mode for 5G-NR tests.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **latency_ cell_primary_ signal_strength** | Integer | -103 | Measured signal strength in dBm of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Received Signal Strength Indication (RSSI) for 3G tests or the Reference Signal Received Power (RSRP) for 4G LTE or 5G-NR tests.* |
| **latency_ cell_primary_ rssi** | Integer | | Measured Received Signal Strength Indication (RSSI) in dBm of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 5G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ rsrp** | Integer | -103 | Measured Reference Signal Received Power (RSRP) in dBm of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |

| Field | Data Type | Example | Description / Notes |
| --- | --- | --- | --- |
| **latency_ cell_primary_ rsrq** | Integer | -12 | Measured Reference Signal Received Quality (RSRQ) in dB of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **latency_ cell_primary_ sinr** | Integer | 2 | Measured Signal to Interference and Noise Ratio (SINR) in dB of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: this value represents the Synchronization Signal (SS) for 5G-NR tests and the Channel-specific Reference Signal (CRS) for 4G LTE tests.* |
| **latency_ cell_primary_ csi_rsrp** | Integer | -103 | Measured 5G Channel State Information (CSI) RSRP in dBm of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ cell_primary_ csi_rsrq** | Integer | -12 | Measured Channel State Information (CSI) RSRQ in dB of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ csi_sinr** | Integer | 2 | Measured Channel State Information (CSI) SINR in dB of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G or 4G LTE tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ cqi** | Integer | 4 | Measured Channel Quality Indicator (CQI) of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value must be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ spectrum_band** | String | n41 | Spectrum band used by the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null for 3G tests.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.*<br>*- Note: the reported band value corresponds to the Operating Bands tables as follows:*<br>*- 4G LTE:  3GPP TS 36.101 section 5.5*<br>*- 5G-NR:  3GPP TS 38.101 table 5.2-1* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **latency_ cell_primary_ spectrum_ bandwidth** | Numeric | 100.0 | Total amount of spectral bandwidth used by the primary cell for the latency test metric in MHz.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |
| **latency_ cell_primary_ arfcn** | Integer | 528000 | Measured physical RF channel number of the primary cell for the latency test metric.<br>*- Value is not available on iOS and may be null for these device types.*<br>*- Value may be null if the device does not return a valid value or else returns a value of unknown.* |

## 6 Entity Mobile Testing Methodology Information

### 6.1 Overview

Mobile broadband service providers, governmental entities, and other third parties must submit a complete description of their data collection methodology when using their own hardware and software to gather on-the-ground speed test data. Parties must collect the required metrics that the FCC Speed Test app and OET-approved apps must gather as set forth above.

Parties must provide a narrative overview of the testing solution, including a description of the test methodology and a URL link to the test solution product (if available). They should also include, if available, product images, sample reports from test campaigns conducted using this product, and any other information that is useful in determining the product test and measurement capabilities.

In addition to the narrative overview, parties must also submit structured information about their test methodologies via a web form in the BDC system, described below.

### 6.2 Entity Mobile Testing Methodology Data

This web form requires filers to enter a description of the testing methodology the used to collect the Entity Mobile Crowdsourced/Challenge Information.

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **Methodology ID** | String | Acme Wireless Drive Test - Team B | Unique name identifying the testing methodology used to collect the mobile speed test data. |
| **Narrative** | File Upload | | Narrative explanation of the methodology used to conduct the mobile speed tests.<br>*- File must be a valid Word Document (.DOC/.DOCX), OpenDocument (.ODT), Portable Document Format (.PDF), or plain text (.TXT) file.* |
| **Category** | Enumerated | App-based | Category of testing methodology.<br>*- Value must be one of the following:*<br>▪ *App-based*<br>▪ *Software-based*<br>▪ *Hardware-based* |
| **Supported Technologies** | Enumerated | 4G LTE<br>5G-NR NSA<br>5G-NR SA | List of radio access technologies supported by the device(s) used for testing.<br>*- Value(s) must be one or more of the following:*<br>▪ *3G*<br>▪ *4G LTE*<br>▪ *5G-NR NSA*<br>▪ *5G-NR SA* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **Testing Environment** | Enumerated | In-vehicle | Testing environment used for the test setup.<br>- *Value must be one of the following:*<br>▪ *In-vehicle*<br>▪ *Outdoors* |
| **Antenna Type** | Enumerated | External | Type of antenna used for the test setup.<br>- *Value must be one of the following:*<br>▪ *External*<br>▪ *Internal* |
| **External Antenna Height** | Decimal | 0.5 | Height of external antenna in meters.<br>- *Value must be ≥ 0 if Antenna Type value is "External".*<br>- *Value must be null if the Antenna Type value is "Internal".* |
| **External Antenna Gain** | Decimal | 5.0 | Gain of external antenna in dB.<br>- *Value must be ≥ 0 if Antenna Type value is "External".*<br>- *Value must be null if the Antenna Type value is "Internal".* |
| **Data Plan Type** | Enumerated | Unlimited (Not Subject to any Data Usage Management) | Type of data plan used by the devices.<br>- *Value must be one of the following:*<br>▪ *Unlimited (Subject to Typical Consumer Data Usage Management)*<br>▪ *Unlimited (Not Subject to any Data Usage Management)*<br>▪ *Other* |
| **Data Plan Description** | String | | Description of data plan used by the device(s) in the test setup.<br>- *Value may be null unless the Data Plan Type value is "Other".* |

### 6.3 Entity Mobile Test Servers Data

This web form requires filers to enter a description of the test servers corresponding to the methodology used to collect the Entity Mobile Crowdsourced/Challenge Information.

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **Server ID** | String | Acme Wireless Stackpath Cloud | Unique name identifying the test server record. |
| **Server Count** | Integer | 15 | Minimum count of test servers available for transmitting data for a speed test. |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **Server Type** | Enumerated | Geographically-distributed / Nationwide Test Servers with Common Configuration | Type of test server(s) for which the test record applies.<br>- *Value must be one of the following:*<br>▪ *Geographically-distributed / Nationwide Test Servers with Common Configuration*<br>▪ *Non-distributed / Region-specific Test Server(s)* |
| **Server Location - State** | Enumerated | | State or territory in which the test servers are located.<br>- *Value must be null if the Server Type value is "Geographically-distributed / Nationwide Test Servers with Common Configuration".*<br>- *Value must be a valid state or territory from the latest U.S. Census Bureau data if not null.* |
| **Server Location - County** | Enumerated | | County or other subdivision within the state or territory in which the test servers are located.<br>- *Value must be null if the Server Type value is "Geographically-distributed / Nationwide Test Servers with Common Configuration".*<br>- *Value must be a valid county or county-level subdivision for the selected state from the latest U.S. Census Bureau data if not null.* |
| **Server Capacity - Incoming** | Decimal | 100.0 | Minimum one-way provisioned capacity of test server link in Gbps given capabilities of existing hardware – incoming to the test server.<br>- *Value must be > 0.* |
| **Server Capacity - Outgoing** | Decimal | 100.0 | Minimum one-way provisioned capacity of test server link in Gbps given capabilities of existing hardware – outgoing from the test server.<br>- *Value must be > 0.* |
| **Server Client - Type** | Enumerated | iPerf | Type of client software running on the test server(s).<br>- *Value must be one of the following:*<br>▪ *iPerf*<br>▪ *Ookla*<br>▪ *Other* |
| **Server Client - Description** | String | | Description of the client type used by the test server(s).<br>- *Value may be null unless the Server Client Type value is "Other".* |

| Field | Data Type | Example | Description / Notes |
|---|---|---|---|
| **Server Client - Protocol** | Enumerated | TCP | Transport layer network protocol used by the client software running on the test server(s).<br>- *Value must be one of the following:*<br>▪ *TCP*<br>▪ *UDP*<br>▪ *SCTP*<br>▪ *Other* |
| **Server Client - Port** | Integer | | Network port used by the client software running on the test server(s).<br>- *Value may be null.* |
| **Server Client - Buffer Length** | Integer | | Buffer length used by the client software running on the test server(s).<br>- *Value may be null.* |
| **Server Client - Max Connections** | Integer | | Maximum number of simultaneous connections allowed by the client software running on the test server(s).<br>- *Value may be null.* |
| **Server Client - Configuration Description** | String | | Description of any other configuration of the client software running on the test server(s).<br>- *Value may be null.* |